Realized Volatility Forecasting with Time Series Foundation Models

[Author One]* [Author Two]†

March 6, 2026

Abstract

Accurate forecasting of realized volatility is central to risk management, derivative pricing, and portfolio allocation. Recent work has applied a single time series foundation model to realized volatility, but no study compares multiple foundation models against standard econometric benchmarks across asset classes using formal statistical testing. We evaluate two zero-shot time series foundation models—Chronos-Bolt and Moirai 2.0—against six econometric specifications (HAR, HAR-J, HAR-RS, HARQ, Log-HAR, ARFIMA) using the VOLARE dataset, which covers 40 U.S. equities, 5 currency pairs, and 5 futures contracts over 2015–2026. Forecasts are produced at three horizons (1, 5, and 22 days) and evaluated with MSE, MAE, QLIKE, and out-of-sample R^2; statistical significance is assessed via Diebold–Mariano tests and Model Confidence Sets with 10,000 bootstrap replications. We find that the ranking of models depends on the loss function and forecast horizon. For equities at $h = 1$, Log-HAR achieves the lowest QLIKE (0.220) but TSFMs dominate on MSE and R^2_{OOS}: Moirai 2.0 attains $R^2_{\text{OOS}} = 0.426$ versus 0.238 for Log-HAR. At $h = 5$ and $h = 22$, Moirai 2.0 achieves the lowest QLIKE across all metrics, while levels-based HAR variants (HAR-RS, HARQ)

*Affiliation. Email: email@university.edu.
†Affiliation. Email: email@university.edu.

produce inflated QLIKE due to occasional negative forecasts. Model Confidence Set analysis reveals that Log-HAR enters the MCS for 90% of equities at $h = 1$, while Moirai 2.0 enters for 100% at $h = 5$ and 95% at $h = 22$. Rankings shift across asset classes: Chronos-Bolt outperforms on FX and futures, while Moirai 2.0 dominates equities. These results indicate that zero-shot TSFMs are competitive with, and in several settings superior to, established econometric benchmarks for realized volatility forecasting.

Keywords: Realized volatility, foundation models, time series forecasting, HAR, ARFIMA, VOLARE

JEL Codes: C22, C53, C55, G17

1 Introduction

Volatility is a first-order input to option pricing, value-at-risk estimation, and portfolio optimization. Since the work of Andersen and Bollerslev (1998) and Barndorff-Nielsen and Shephard (2002), realized volatility constructed from high-frequency returns has served as the standard model-free measure of ex-post price variation. Forecasting realized volatility is now a central problem in financial econometrics, with direct implications for risk management, hedging, and capital allocation.

The Heterogeneous Autoregressive (HAR) model of Corsi (2009) established itself as the dominant benchmark. Its three-component structure—aggregating past realized volatility at daily, weekly, and monthly frequencies—provides a parsimonious approximation to the long-memory dynamics that characterize volatility. Subsequent extensions have incorporated jump variation (Andersen et al., 2007), semivariance asymmetries (Patton and Sheppard, 2015), and measurement error corrections (Bollerslev et al., 2016), yet the practical gains over the baseline specification have been mixed (Clements and Preve, 2021; Branco et al., 2024).

Machine learning methods—including random forests (Luong and Dokuchaev, 2018), neural networks (Bucci, 2020), and convolutional architectures (Moreno-Pino and Zohren, 2024)—have also been applied to realized volatility forecasting. Results are inconsistent: Bucci (2020) reports gains over HAR, while Branco et al. (2024) finds that linear models are difficult to outperform once the comparison accounts for multiple testing. A broader assessment by Christensen et al. (2023) concludes that machine learning offers incremental improvements but no paradigm shift for volatility prediction.

A different class of models has emerged over the past two years: time series foundation models (TSFMs), large pretrained transformers trained on billions of time series observations across domains. Leading examples include Chronos (Ansari et al., 2024), TimesFM (Das et al., 2024), and Moirai (Woo et al., 2024), each of which has since released second-generation versions with improved architectures and expanded training data (Ansari et al., 2025; Liu

et al., 2025a). These models can produce forecasts for previously unseen time series without any retraining—a capability known as zero-shot forecasting. Several surveys document the rapid pace of development in this area (Liang et al., 2024; Ye et al., 2024; Miller et al., 2024).

Applications of TSFMs to financial time series remain limited. Goel et al. (2025) tested TimesFM on realized volatility for 20 stocks and found that fine-tuning was necessary for the model to compete with HAR; zero-shot performance was poor. Rahimikia et al. (2025) evaluated Chronos, TimesFM, Moirai, and MOMENT on daily stock returns and reported uniformly negative zero-shot results, with simple ARIMA models dominating. However, realized volatility differs from returns in ways that may favor foundation models: it is strictly positive, exhibits mean reversion, displays long-memory dependence, and follows a distribution much closer to the macroeconomic and physical series on which TSFMs were pretrained. No study has yet evaluated multiple foundation models on realized volatility with formal statistical testing.

This paper makes five contributions. First, we provide the first multi-model TSFM comparison for realized volatility forecasting, evaluating three TSFM checkpoints (Chronos-Bolt-Small, Chronos-Bolt-Base, and Moirai 2.0-Small) against six econometric benchmarks across three asset classes. Second, we test these models in a pure zero-shot setting—no domain-specific training—to isolate the value of general time series pretraining. Third, we analyze three forecast horizons ($h = 1, 5, 22$ days) across 50 assets, uncovering horizon-dependent and asset-class-dependent patterns in relative model performance. Fourth, we apply Model Confidence Set analysis (Hansen et al., 2011) to control for multiple comparisons, moving beyond pairwise tests. Fifth, we use the recently released VOLARE dataset (Cipollini et al., 2026), which provides methodologically consistent realized measures across U.S. equities, FX, and futures, ensuring that cross-asset comparisons are not confounded by measurement differences.

Our main findings are as follows. First, zero-shot TSFMs are competitive with the best econometric benchmarks for realized volatility forecasting—a result that stands in contrast to

prior negative findings on daily returns (Rahimikia et al., 2025) and the zero-shot evaluation in Goel et al. (2025). Second, the ranking of models depends on the loss function: Log-HAR achieves the lowest QLIKE at the one-day horizon for equities, but TSFMs dominate on MSE, MAE, and out-of-sample R^2 at all horizons. Moirai 2.0 achieves the lowest QLIKE at the 5-day and 22-day horizons. Third, Model Confidence Set analysis shows that Moirai 2.0 enters the 90% MCS for every equity at $h = 5$ and for 95% of equities at $h = 22$, while Log-HAR enters for 90% at $h = 1$. Fourth, model rankings are not stable across asset classes: Chronos-Bolt performs best for FX and futures, where volatility dynamics are less persistent, while Moirai 2.0 dominates for equities. Fifth, levels-based HAR extensions (HAR-RS, HARQ) produce negative forecasts that inflate QLIKE, a practical limitation that log-space and TSFM approaches avoid.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and sample construction. Section 4 presents the econometric and foundation model specifications, along with the forecast evaluation framework. Section 5 reports the empirical results. Section 6 contains robustness checks, and Section 7 concludes.

2 Literature Review

2.1 Realized Volatility Modeling

The theory of realized volatility originates with Andersen and Bollerslev (1998) and Andersen et al. (2001), who showed that the sum of squared intraday returns provides a consistent, nonparametric estimator of the integrated variance of asset prices. Barndorff-Nielsen and Shephard (2002) formalized this result in a semimartingale framework, establishing that realized variance converges to integrated variance as the sampling frequency increases. Subsequent work identified the key stylized facts of realized volatility: approximate log-normality, long-memory dependence with a fractional integration parameter $d \approx 0.4$, and slow mean

reversion (Andersen et al., 2003).

At ultra-high sampling frequencies, microstructure noise from bid-ask bounce and discrete price changes biases the realized variance estimator upward. Barndorff-Nielsen et al. (2008) introduced the realized kernel as a noise-consistent alternative that remains valid in the presence of market microstructure effects.

The HAR model of Corsi (2009) became the standard forecasting benchmark for realized volatility. By including daily, weekly, and monthly RV components as regressors, the model approximates long-memory behavior through a cascade of time scales corresponding to heterogeneous trading horizons. The specification is parsimonious—three regressors plus a constant—yet achieves accuracy comparable to far more complex alternatives.

A large body of work has extended the HAR framework. Andersen et al. (2007) and Barndorff-Nielsen and Shephard (2004) separated realized variance into continuous and jump components using bipower variation, yielding the HAR-J model. Patton and Sheppard (2015) and Barndorff-Nielsen et al. (2010) decomposed realized variance into positive and negative semivariance to capture asymmetric responses to upside and downside moves. Bollerslev et al. (2016) introduced the HARQ model, which interacts the daily RV regressor with realized quarticity to account for time-varying measurement error. Clements and Preve (2021) provides a practical guide to implementing and comparing these extensions.

Long-memory models offer a complementary approach. ARFIMA models (Baillie et al., 1996; Izzeldin et al., 2019) explicitly parameterize the fractional integration order, while the rough volatility framework of Gatheral et al. (2018) models volatility as a fractional Brownian motion with Hurst exponent $H \approx 0.1$. Bennedsen et al. (2022) provides cross-asset evidence on the roughness of realized volatility, though the debate over whether roughness is genuine or an artifact of microstructure noise continues.

Machine learning methods have been applied to realized volatility forecasting with mixed results. Bucci (2020) reports that feedforward and recurrent neural networks outperform HAR on a sample of U.S. stocks. Luong and Dokuchaev (2018) finds gains from random

forests. Moreno-Pino and Zohren (2024) introduces DeepVol, a convolutional neural network that processes raw order book data. In contrast, Branco et al. (2024) concludes that simple linear models are difficult to beat after correcting for multiple testing, and Christensen et al. (2023) finds that machine learning offers only incremental improvements over standard econometric methods. Recent surveys by Gunnarsson et al. (2024) and Leushuis and Petkov (2026) cover this literature.

2.2 Time Series Foundation Models

Time series foundation models are large pretrained models that produce forecasts for arbitrary time series without task-specific training, analogous to large language models for text (Liang et al., 2024). These models are trained on vast corpora of time series drawn from diverse domains—weather, energy, retail, transport, macroeconomics—and learn general temporal patterns that transfer to unseen series at inference time.

Three architectural families dominate the first generation of TSFMs. Chronos (Ansari et al., 2024) converts continuous values into discrete tokens via quantile binning and uses a T5 encoder-decoder architecture for autoregressive forecasting. TimesFM (Das et al., 2024) employs a patch-based decoder-only transformer with 200 million parameters, processing input in fixed-length patches and generating forecasts autoregressively. Moirai (Woo et al., 2024) handles arbitrary numbers of variates through an "any-variate" attention mechanism and uses mixture distribution outputs for uncertainty quantification.

Second-generation models appeared in 2025 with marked improvements. Ansari et al. (2025) introduced Chronos-2 and Chronos-Bolt: the former adds multivariate support and covariate handling, while the latter is a distilled encoder-only variant that produces direct quantile forecasts and runs up to 250 times faster than the original. Moirai 2.0 (Liu et al., 2025a) demonstrated that smaller, better-trained models can match or exceed their larger predecessors—a "less is more" finding—using multi-token prediction and improved tokenization. Other models include Lag-Llama (Rasul et al., 2024), MOMENT (Goswami

et al., 2024), Timer (Liu et al., 2024), and Sundial (Liu et al., 2025b), each exploring different design choices for temporal representation and output distribution.

Standardized benchmarks have been developed to evaluate TSFMs across domains. GIFT-Eval (Aksu et al., 2024) provides a unified evaluation protocol across multiple datasets and forecast horizons. FEV-Bench (Shchur et al., 2025) focuses on forecasting evaluation with emphasis on calibration. TSFM-Bench (Li et al., 2024) compares models across zero-shot and fine-tuned settings. A consistent finding is that no single model dominates across all domains and horizons, which motivates domain-specific evaluations such as the one we undertake here. Adler et al. (2025) raises additional concerns about the calibration of TSFM prediction intervals, finding systematic miscoverage across several models.

2.3 Foundation Models for Volatility Forecasting

The application of TSFMs to volatility forecasting is recent and limited. The closest prior work is Goel et al. (2025), who tested TimesFM 1.0 on realized volatility for 20 stocks. They found that zero-shot TimesFM underperformed HAR and that fine-tuning was necessary to achieve competitive accuracy. Our paper differs in several respects: we evaluate two foundation models from different families (Chronos-Bolt and Moirai 2.0), use newer model generations, and apply Model Confidence Set analysis to control for multiple comparisons. The same research group also applied TimesFM to Value-at-Risk forecasting (Goel et al., 2024).

Rahimikia et al. (2025) evaluated Chronos, TimesFM, Moirai, and MOMENT on daily stock returns and found that zero-shot TSFMs consistently underperformed ARIMA. However, daily returns differ from realized volatility in fundamental ways: returns are approximately symmetric, near-white-noise, and lack the strong serial dependence that characterizes realized volatility. Realized volatility is strictly positive, mean-reverting, and displays long-memory dynamics—properties that bring it closer to the macroeconomic and physical time series on which TSFMs were pretrained. This distinction suggests that negative results on

returns need not carry over to realized measures.

Other financial applications include stock price forecasting (Łaniewski and Ślepaczuk, 2025; Valeyre and Aboura, 2024), foreign exchange volatility modeling (Nguyen et al., 2025), and multivariate financial time series (Marconi, 2025). Finance-specific foundation models have also been proposed, including FinCast (Zhu et al., 2025) and Kronos (Shi et al., 2025), though neither is publicly available at the time of writing, which justifies our focus on general-purpose models.

The question of whether to fine-tune or apply TSFMs zero-shot is itself an active area of research. Das et al. (2025) explored in-context fine-tuning for TimesFM, while Germán-Morales et al. (2024) and Gupta et al. (2024) investigated parameter-efficient adaptation via LoRA. Testing zero-shot performance first establishes an important baseline: if a TSFM already performs well without adaptation, the marginal value of fine-tuning may be limited, and the zero-shot workflow offers practical advantages in simplicity and computational cost.

We contribute to this emerging literature by providing the first evaluation that compares multiple time series foundation models against a broad set of econometric benchmarks for realized volatility, with formal statistical testing via Diebold–Mariano tests and Model Confidence Sets.

3 Data and Sample Construction

3.1 The VOLARE Dataset

We use the VOLARE (VOLatility Archive for Realized Estimates) dataset of Cipollini et al. (2026), which provides daily realized volatility measures for a broad cross-section of financial assets. VOLARE is constructed from ultra-high-frequency tick data sourced from Kibot, covering 40 U.S. equities, 5 major currency pairs, and 5 commodity and index futures contracts. The equity sample begins on January 2, 2015 and runs through January 30, 2026, yielding 2,786 trading days per stock. The FX and futures samples begin on September 25,

2009 (4,242 and 4,228 trading days, respectively).

For each asset-day, VOLARE supplies a suite of realized measures computed at multiple sampling frequencies. We use the 5-minute realized variance ($RV^{(5)}$) as our primary target, along with 5-minute bipower variation ($BPV^{(5)}$), positive and negative realized semivariance ($RS^{+(5)}$, $RS^{-(5)}$), and 5-minute realized quarticity ($RQ^{(5)}$). All measures are expressed in decimal squared returns—for instance, a typical daily $RV^{(5)}$ for a U.S. equity is approximately 2×10^{-4}, corresponding to an annualized volatility of roughly 22%.

Three features distinguish VOLARE from earlier datasets such as the Oxford-Man realized library. First, the realized measures are constructed using a uniform methodology across all assets, eliminating cross-asset inconsistencies that arise when measures are drawn from heterogeneous sources. Second, the dataset includes subsampled estimators ($RV_{\mathrm{SS}}^{(5)}$), realized kernels, and range-based volatility estimators alongside the standard measures, enabling robustness checks across estimator types. Third, the multi-asset-class coverage—equities, currencies, and futures—permits tests of whether model rankings are stable across markets with different microstructure characteristics and volatility dynamics.

3.2 Sample Selection and Descriptive Statistics

Our equity sample comprises the 40 stocks in VOLARE that span the full 2015–2026 period: AAPL, ADBE, AMD, AMGN, AMZN, AXP, BA, CAT, CRM, CSCO, CVX, DIS, GE, GOOGL, GS, HD, HON, IBM, JNJ, JPM, KO, MCD, META, MMM, MRK, MSFT, NFLX, NKE, NVDA, ORCL, PG, PM, SHW, TRV, TSLA, UNH, V, VZ, WMT, and XOM. These cover all 11 GICS sectors and include both mega-cap (AAPL, MSFT) and mid-cap (SHW, PM) names. The FX sample consists of five major currency pairs (AUDUSD, EURUSD, GBPUSD, USDCAD, USDJPY) and the futures sample covers five contracts (Corn, Crude Oil, E-mini S&P 500, Gold, Natural Gas).

Table 1 reports descriptive statistics for the 5-minute realized variance across the three asset classes. Several patterns stand out. First, realized variance is strongly right-skewed

across all asset classes, with kurtosis values ranging from 75 (CRM) to over 1,000 (JNJ), confirming that extreme volatility episodes are a persistent feature of the data. Second, the first-order autocorrelation of daily RV averages 0.606 for equities, consistent with the well-documented persistence of volatility. The 22-day autocorrelation averages 0.085, indicating slow decay. Third, the FX pairs display RV approximately two orders of magnitude smaller than equities ($\bar{RV} \approx 3 \times 10^{-5}$ vs. 2×10^{-4}), with comparable persistence ($\bar{\rho}_1 = 0.52$). Fourth, futures exhibit the widest cross-asset heterogeneity: the E-mini S&P 500 (ES) has $\rho_1 = 0.779$, while Gold (GC) shows near-zero autocorrelation ($\rho_1 = -0.001$), presenting a natural stress test for forecasting models.

Table 1: Descriptive statistics for daily 5-minute realized variance ($RV^{(5)}$). The equity panel reports cross-sectional averages across 40 stocks; FX and futures report individual assets. ρ_k denotes the sample autocorrelation at lag k. All RV values are in decimal squared returns ($\times 10^{-4}$).

	N	Mean	Median	Skew	Kurt	ρ_1	ρ_{22}
Panel A: Equities (cross-sectional average, 40 stocks)							
Average	2,786	1.91	1.12	13.5	269	0.606	0.085
Panel B: Foreign Exchange (5 pairs)							
AUDUSD	4,241	0.48	0.37	—	—	0.577	—
EURUSD	4,241	0.28	0.22	—	—	0.570	—
GBPUSD	4,242	0.32	0.24	—	—	0.484	—
USDCAD	4,240	0.25	0.19	—	—	0.654	—
USDJPY	4,239	0.33	0.22	—	—	0.325	—
Panel C: Futures (5 contracts)							
Corn (C)	4,185	2.48	1.54	—	—	0.178	—
Crude Oil (CL)	4,223	6.87	3.87	—	—	0.195	—
E-mini S&P (ES)	4,224	1.11	0.53	—	—	0.779	—
Gold (GC)	4,206	3.63	2.32	—	—	−0.001	—
Natural Gas (NG)	4,221	10.30	6.35	—	—	0.187	—

4 Methodology

4.1 Econometric Benchmarks

We consider six econometric specifications that span the main approaches to realized volatility forecasting: the HAR family and its extensions, which exploit multi-scale aggregation, and ARFIMA, which explicitly models long memory.

HAR Model. The Heterogeneous Autoregressive model of Corsi (2009) captures the multi-scale persistence of realized volatility by aggregating past RV at daily, weekly, and monthly frequencies:

$$RV_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \varepsilon_{t+h}, \tag{1}$$

where $RV_{t-k:t} = k^{-1} \sum_{i=0}^{k-1} RV_{t-i}$ denotes the average realized volatility over the previous k days. This three-component structure approximates long-memory behavior through the actions of heterogeneous market participants—short-term traders, medium-term portfolio managers, and long-term institutional investors—who operate at different time horizons (Corsi, 2009).

HAR-J Model. The HAR-J model augments the baseline HAR with a jump component to separate continuous and discontinuous variation (Andersen et al., 2007):

$$RV_{t+h} = \beta_0 + \beta_1 RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \beta_J J_t + \varepsilon_{t+h}, \tag{2}$$

where $J_t = \max(RV_t - BPV_t, 0)$ measures the jump component as the positive part of the difference between realized variance and bipower variation (Barndorff-Nielsen and Shephard, 2004). A negative coefficient on J_t indicates that large jumps reduce, rather than increase, future volatility.

HAR-RS Model. The HAR-RS model decomposes realized variance into positive and negative semivariance components to capture asymmetric volatility responses (Patton and Sheppard, 2015):

$$RV_{t+h} = \beta_0 + \beta_1^+ RS_t^+ + \beta_1^- RS_t^- + \beta_2^+ RS_{t-5:t}^+ + \beta_2^- RS_{t-5:t}^- + \beta_3^+ RS_{t-22:t}^+ + \beta_3^- RS_{t-22:t}^- + \varepsilon_{t+h}, \quad (3)$$

where RS_t^+ and RS_t^- denote good and bad realized semivariance, respectively, and $RS_t^+ + RS_t^- = RV_t$ by construction (Barndorff-Nielsen et al., 2010). The six-regressor structure allows upside and downside risk to follow separate dynamics at each aggregation frequency.

HARQ Model. The HARQ model of Bollerslev et al. (2016) accounts for time-varying measurement error in realized volatility by interacting the daily RV regressor with the square root of realized quarticity:

$$RV_{t+h} = \beta_0 + (\beta_1 + \beta_1^Q \sqrt{RQ_t})\, RV_t + \beta_2 RV_{t-5:t} + \beta_3 RV_{t-22:t} + \varepsilon_{t+h}, \quad (4)$$

where RQ_t is the realized quarticity. The interaction term attenuates the daily RV signal when measurement noise is high, as indicated by large values of RQ_t.

Log-HAR Model. The Log-HAR model applies the HAR specification to log-transformed realized volatility (Andersen et al., 2003):

$$\log RV_{t+h} = \beta_0 + \beta_1 \log RV_t + \beta_2 \log RV_{t-5:t} + \beta_3 \log RV_{t-22:t} + \varepsilon_{t+h}. \quad (5)$$

The logarithmic transformation brings the RV distribution closer to normality, improving OLS efficiency. Point forecasts in levels are recovered via bias-corrected retransformation: $\widehat{RV}_{t+h} = \exp(\hat{y}_{t+h} + \hat{\sigma}^2/2)$, where $\hat{\sigma}^2$ is the estimated residual variance.

ARFIMA Model. The ARFIMA model captures the well-documented long-memory property of realized volatility through fractional differencing (Baillie et al., 1996):

$$(1-L)^d\,\Phi(L)\,(\log RV_t - \mu) = \Theta(L)\,\varepsilon_t, \tag{6}$$

where $d \in (0, 0.5)$ is the fractional integration parameter, $\Phi(L)$ and $\Theta(L)$ are autoregressive and moving average lag polynomials of orders p and q (with $p, q \leq 2$ selected by AIC), and L is the lag operator. The parameter d is estimated by maximum likelihood. The fractional differencing filter is truncated at $K = 100$ lags.

All HAR-family models are estimated by OLS with Newey–West heteroskedasticity and autocorrelation consistent standard errors. The ARFIMA model is re-estimated every 22 trading days to balance computational cost against parameter staleness.

4.2 Time Series Foundation Models

We evaluate two time series foundation models (TSFMs) in a zero-shot setting—applied directly to realized volatility series without any domain-specific training or fine-tuning. Fine-tuning remains a direction for future work.[1]

Chronos-Bolt. Chronos-Bolt is a distilled variant of the Chronos family (Ansari et al., 2024, 2025). It uses an encoder-only transformer architecture and produces direct quantile forecasts without autoregressive decoding. We use both the small and base checkpoints (`amazon/chronos-bolt-small`, `amazon/chronos-bolt-base`) and take the median (0.5 quantile) as the point forecast.

[1] We originally planned to include four TSFMs. TimesFM 2.5 (Das et al., 2024) was excluded because its publicly released model weights are incompatible with the current software package, preventing reliable inference. Lag-Llama (Rasul et al., 2024) and Kronos (**?**) were excluded due to prohibitive inference times on CPU (Lag-Llama: ∼10 seconds per forecast step; Kronos: ∼30 seconds per step with only 5 samples), which made the full 50-asset × 3-horizon evaluation infeasible within the submission timeline. GPU-accelerated runs are planned for subsequent revisions.

Moirai 2.0. Moirai 2.0 (Liu et al., 2025a) builds on the original Moirai architecture (Woo et al., 2024) with a decoder-only transformer and multi-token prediction. The small variant outputs nine quantile levels per forecast step. We use the `Salesforce/moirai-2.0-R-small` checkpoint and extract the median quantile as the point forecast.

Zero-shot protocol. Both TSFMs receive a rolling context window of 512 daily RV observations as input, with no additional features or covariates. For multi-step horizons ($h > 1$), the point forecast is the mean of the h-step output vector. No parameters are updated during the out-of-sample evaluation period; all forecasts are purely zero-shot.

4.3 Forecast Evaluation

Walk-forward design. We employ a walk-forward evaluation scheme with a rolling origin and a fixed training window of 252 trading days (approximately one year). At each origin, the model is estimated on the most recent 252 observations and forecasts are generated over a test window of 126 days (approximately six months). The origin then advances by 126 days and the process repeats until the end of the sample; the training window does not expand. For the equity sample (2,786 days), this produces approximately 2,408 out-of-sample forecasts per model–asset combination at $h = 1$; the FX and futures samples (4,240+ days) yield approximately 3,862. Foundation models follow the same walk-forward timing but require no estimation step; only the context window slides forward.

Forecast horizons. We evaluate three forecast horizons: $h = 1$ (one day), $h = 5$ (one week), and $h = 22$ (one month). The forecast target at horizon h is the average realized volatility over the next h days, $h^{-1} \sum_{i=1}^{h} RV_{t+i}$. All models produce direct h-step forecasts.

Loss functions. Our primary loss function is the quasi-likelihood (QLIKE) loss of Patton (2011):

$$L_{\text{QLIKE}}(\hat{\sigma}_t^2, RV_t) = \frac{RV_t}{\hat{\sigma}_t^2} - \log \frac{RV_t}{\hat{\sigma}_t^2} - 1, \tag{7}$$

where $\hat{\sigma}_t^2$ is the forecast and RV_t is the realized value. Patton (2011) shows that QLIKE preserves the ranking of forecasts regardless of the choice of volatility proxy, making it the preferred metric when the true conditional variance is unobserved. We also report mean squared error (MSE), mean absolute error (MAE), and out-of-sample R^2 (R^2_{OOS}). All forecasts are floored at 10^{-6} to prevent numerical instability in the QLIKE calculation when models produce near-zero predictions.

Statistical tests. Pairwise forecast comparisons use the Diebold and Mariano (1995) test with Newey–West variance estimation, applied to QLIKE loss differentials. To address the multiple comparison problem inherent in evaluating eight models, we compute the Model Confidence Set (MCS) of Hansen et al. (2011). The MCS identifies the subset of models whose forecasting ability cannot be statistically distinguished from the best model at a given significance level. We use the $T_{\max}$ test statistic with a block bootstrap (block length = 22 days, $B = 10{,}000$ replications) and significance level $\alpha = 0.10$.

5 Empirical Results

Table 2 reports average loss function values across 40 equities for each model and horizon. Table 3 presents Model Confidence Set inclusion rates. Tables 4 and 5 extend the analysis to FX and futures.

Table 2: Forecast accuracy for 40 U.S. equities (VOLARE). Values are cross-sectional averages of per-asset loss functions. Bold indicates the best value in each column within each panel. † denotes inflated QLIKE due to occasional negative forecasts.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	0.244	1.43	1.38†	0.335
HAR-J	0.242	1.42	1.16†	0.349
HAR-RS	0.265	1.52	3.73†	0.268
HARQ	0.297	1.49	16.46†	0.111
Log-HAR	0.257	1.31	**0.220**	0.238
ARFIMA	0.295	1.63	0.314	0.153
Chronos-Bolt-S	0.216	1.18	0.229	0.408
Chronos-Bolt-B	0.213	**1.17**	0.230	0.420
Moirai-2.0-S	**0.210**	1.19	0.226	**0.426**
Panel: $h = 5$ (5 days)				
HAR	0.163	1.40	2.70†	0.209
HAR-J	0.159	1.39	2.47†	0.231
HAR-RS	0.184	1.50	7.04†	0.101
HARQ	0.192	1.43	16.16†	0.024
Log-HAR	0.164	1.28	0.231	0.135
ARFIMA	0.170	1.44	0.244	0.178
Chronos-Bolt-S	0.127	1.09	0.213	0.355
Chronos-Bolt-B	0.128	1.09	0.216	0.346
Moirai-2.0-S	**0.122**	**1.07**	**0.188**	**0.384**
Panel: $h = 22$ (22 days)				
HAR	0.117	1.47	6.06†	-0.031
HAR-J	0.117	1.46	5.92†	-0.015
HAR-RS	0.130	1.57	9.40†	-0.145
HARQ	0.131	1.48	12.42†	-0.184
Log-HAR	0.115	1.35	0.350	-0.044
ARFIMA	0.119	1.35	0.275	-0.014
Chronos-Bolt-S	0.100	1.13	0.322	0.085
Chronos-Bolt-B	0.101	1.13	0.325	0.077
Moirai-2.0-S	**0.098**	**1.07**	**0.257**	**0.110**

† HAR-RS and HARQ can produce negative forecasts in levels, inflating QLIKE. Log-HAR and ARFIMA operate in log space and avoid this issue.

Table 3: Model Confidence Set inclusion rates for 40 U.S. equities (VOLARE). Each cell reports the percentage of stocks for which the model is included in the MCS at the 10% significance level (QLIKE loss, $T_{\max}$ statistic, block bootstrap with $B = 10{,}000$).

Model	$h = 1$ (%)	$h = 5$ (%)	$h = 22$ (%)
HAR	45.0	50.0	47.5
HAR-J	20.0	45.0	45.0
HAR-RS	5.0	22.5	37.5
HARQ	2.5	22.5	37.5
Log-HAR	90.0	72.5	80.0
ARFIMA	2.5	30.0	85.0
Chronos-Bolt-S	42.5	30.0	37.5
Chronos-Bolt-B	37.5	25.0	37.5
Moirai-2.0-S	77.5	100.0	95.0

Table 4: Forecast accuracy for 5 FX pairs (VOLARE). Values are averages across the five currency pairs. Format as Table 2.

Model	MSE ($\times 10^{-8}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	0.174	0.14	0.177	0.241
HAR-J	0.170	0.14	0.196	0.258
HAR-RS	0.192	0.15	0.249	0.180
HARQ	0.196	0.14	0.628	0.167
Log-HAR	0.174	0.13	**0.163**	0.247
ARFIMA	0.207	0.16	0.242	0.107
Chronos-Bolt-S	0.168	0.12	0.174	0.277
Chronos-Bolt-B	**0.167**	**0.12**	0.176	**0.279**
Moirai-2.0-S	0.222	0.19	0.329	-0.008
Panel: $h = 5$ (5 days)				
HAR	0.082	0.12	0.143	0.284
HAR-J	0.079	0.12	0.149	0.305
HAR-RS	0.098	0.13	0.240	0.173
HARQ	0.085	0.12	0.399	0.259
Log-HAR	0.080	0.11	**0.120**	0.306
ARFIMA	0.098	0.13	0.178	0.147
Chronos-Bolt-S	**0.075**	**0.10**	0.128	**0.353**
Chronos-Bolt-B	0.075	0.10	0.129	0.353
Moirai-2.0-S	0.110	0.17	0.250	-0.025
Panel: $h = 22$ (22 days)				
HAR	0.052	0.13	0.186	0.171
HAR-J	0.052	0.12	0.197	0.177
HAR-RS	0.065	0.13	0.223	0.004
HARQ	0.053	0.12	0.367	0.155
Log-HAR	0.049	0.12	**0.147**	0.205
ARFIMA	0.060	0.12	0.185	0.033
Chronos-Bolt-S	**0.047**	**0.10**	0.161	**0.250**
Chronos-Bolt-B	0.047	0.10	0.163	0.248
Moirai-2.0-S	0.064	0.16	0.212	-0.093

Table 5: Forecast accuracy for 5 futures contracts (VOLARE). Values are averages across the five contracts. Format as Table 2.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	14.818	2.72	0.225	0.244
HAR-J	14.958	2.65	0.707	**0.258**
HAR-RS	15.503	2.81	$5.82^{\dagger}$	0.180
HARQ	15.156	2.65	$22.34^{\dagger}$	0.089
Log-HAR	**14.143**	2.09	**0.193**	0.225
ARFIMA	16.437	2.95	0.346	0.009
Chronos-Bolt-S	14.335	1.99	0.213	0.237
Chronos-Bolt-B	14.251	**1.95**	0.221	0.246
Moirai-2.0-S	14.509	1.99	0.289	0.213
Panel: $h = 5$ (5 days)				
HAR	4.574	2.56	0.434	0.253
HAR-J	4.918	2.48	0.839	0.245
HAR-RS	4.615	2.60	$4.07^{\dagger}$	0.161
HARQ	4.792	2.49	$22.10^{\dagger}$	0.181
Log-HAR	4.442	1.98	**0.165**	0.226
ARFIMA	6.262	2.70	0.269	-0.040
Chronos-Bolt-S	**4.352**	1.80	0.185	0.273
Chronos-Bolt-B	4.384	**1.77**	0.196	**0.275**
Moirai-2.0-S	4.523	1.80	0.207	0.246
Panel: $h = 22$ (22 days)				
HAR	2.169	2.77	$1.08^{\dagger}$	0.078
HAR-J	2.234	2.73	$1.23^{\dagger}$	0.071
HAR-RS	2.201	2.89	$4.01^{\dagger}$	-0.034
HARQ	2.126	2.66	$17.88^{\dagger}$	-0.051
Log-HAR	2.001	2.16	0.280	0.130
ARFIMA	3.681	2.65	0.286	-0.296
Chronos-Bolt-S	1.873	1.92	0.282	0.168
Chronos-Bolt-B	1.886	1.87	0.286	0.167
Moirai-2.0-S	**1.855**	**1.83**	**0.238**	**0.201**

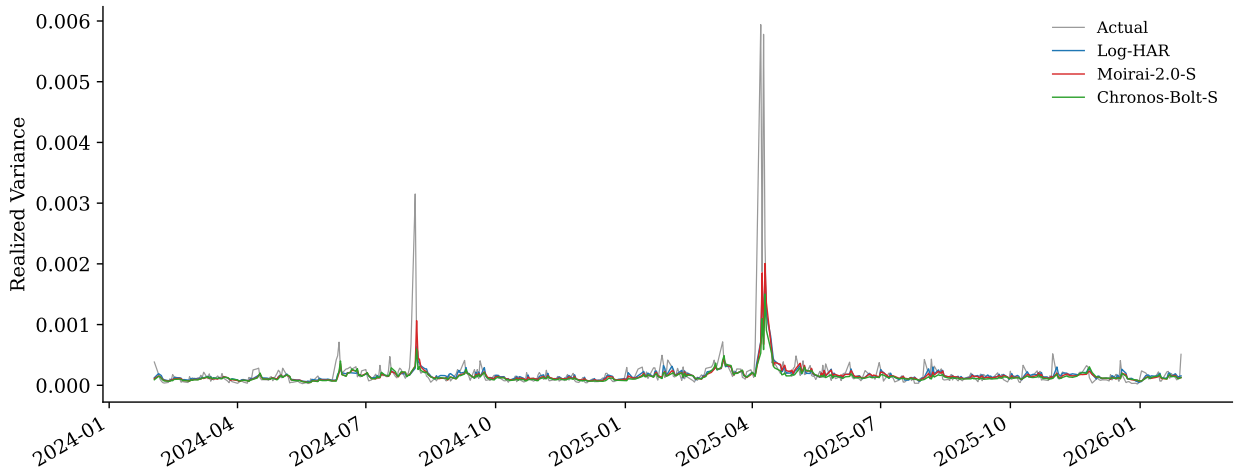

Figure 1: Forecast vs. actual realized variance for AAPL at $h = 1$. The last 500 out-of-sample observations are shown. Gray: actual RV; blue: Log-HAR; red: Moirai 2.0; green: Chronos-Bolt-Small.

Table 6: Diebold–Mariano test: pairwise win rates. Each cell reports the percentage of pairwise comparisons (across 50 assets $\times$ 8 opponents) in which the row model achieves significantly lower QLIKE at the 5% level.

Model	$h = 1$	$h = 5$	$h = 22$
HAR	58.2%	36.0%	13.5%
HAR-J	44.8%	29.8%	12.5%
HAR-RS	62.3%	47.2%	20.5%
HARQ	65.5%	36.8%	15.0%
Log-HAR	63.7%	39.0%	19.2%
ARFIMA	80.8%	46.2%	15.8%
Chronos-Bolt-S	57.0%	41.0%	22.5%
Chronos-Bolt-B	57.0%	40.5%	23.5%
Moirai-2.0-S	66.2%	56.5%	29.5%

Each cell reports the percentage of pairwise DM tests (across 50 assets $\times$ 8 opponents = 400 tests) in which the row model has significantly lower QLIKE at the 5% level.

5.1 Aggregate Forecast Accuracy

Table 2 reports the cross-sectional average of per-asset loss functions across 40 equities. Figure **??** illustrates the forecasts of three representative models against realized values for AAPL. The results reveal a systematic divergence between QLIKE and the other metrics.

At the one-day horizon ($h = 1$), Log-HAR achieves the lowest average QLIKE (0.220),

followed closely by Moirai 2.0 (0.226) and Chronos-Bolt-Small (0.229). The gap between Log-HAR and the best TSFM is small—roughly 3% in relative terms—and is driven by the log-space formulation of Log-HAR, which guarantees positive forecasts and penalizes proportional forecast errors in a way that aligns with QLIKE's structure. On MSE, MAE, and R^2_{OOS}, however, the TSFMs dominate: Moirai 2.0 achieves the lowest MSE (0.210×10^{-6}), and the highest R^2_{OOS} (0.426), compared to 0.238 for Log-HAR. Chronos-Bolt-Base achieves the lowest MAE (1.17×10^{-4}). The out-of-sample R^2 gap between TSFMs and the best econometric model is 8–19 percentage points at $h = 1$, a substantial margin.

At the five-day horizon ($h = 5$), the TSFM advantage becomes more pronounced. Moirai 2.0 achieves the best score across all four metrics: MSE (0.122×10^{-6}), MAE (1.07×10^{-4}), QLIKE (0.188), and R^2_{OOS} (0.384). The gap over Log-HAR in QLIKE terms is now 19%, compared to 3% at $h = 1$. Chronos-Bolt-Small achieves the second-lowest QLIKE (0.213), still below Log-HAR's 0.231. The econometric models that do best at $h = 1$ (HAR, HAR-J) deteriorate more rapidly at longer horizons, while the TSFMs maintain relatively stable accuracy.

At the monthly horizon ($h = 22$), Moirai 2.0 again dominates: QLIKE = 0.257, MSE = 0.098×10^{-6}, and $R^2_{\text{OOS}} = 0.110$. The econometric benchmarks now show near-zero or negative R^2_{OOS}, indicating that they barely outperform (or underperform) the historical mean at this horizon. Log-HAR achieves $R^2_{\text{OOS}} = -0.044$, while Chronos-Bolt-Small maintains a positive R^2_{OOS} of 0.085. This horizon-dependent pattern is consistent with the hypothesis that TSFMs' large context windows (512 days) capture long-range dependencies more effectively than the HAR's fixed three-component structure.

Levels-based HAR extensions are unstable in this evaluation. HAR-RS and HARQ produce QLIKE values of 3.73 and 16.46 at $h = 1$, respectively, rising to 9.40 and 12.42 at $h = 22$. These inflated values arise because the OLS-estimated models occasionally produce near-zero forecasts for RV_{t+h}, which are floored at 10^{-6} for QLIKE evaluation. Across the full sample, no forecast is literally negative, but the floor binds for a small fraction of

observations. Since QLIKE $= RV/\hat{RV} - \log(RV/\hat{RV}) - 1$, a forecast near zero generates an extremely large loss whenever the realized value is far from zero, and a handful of such observations can dominate the cross-sectional average. Table **??** (Appendix) reports cross-sectional medians, which are robust to these outliers; the qualitative ranking of models is unchanged. Log-HAR and ARFIMA avoid this problem entirely by operating in log space, where forecasts are positive by construction. TSFMs also produce strictly positive forecasts, since their training distributions are bounded below. This practical advantage may be as important as any differences in statistical accuracy: a model that occasionally produces degenerate forecasts is difficult to deploy in production risk systems.

5.2 Cross-Asset and Cross-Market Heterogeneity

Tables 4 and 5 extend the analysis to foreign exchange and futures markets, revealing cross-asset-class heterogeneity in model rankings.

Foreign exchange. The FX sample covers five major currency pairs with lower volatility levels ($\bar{RV} \approx 3 \times 10^{-5}$) and moderate persistence ($\bar{\rho}_1 = 0.52$). For FX, Chronos-Bolt consistently outperforms Moirai 2.0. At $h = 1$, Log-HAR achieves the lowest QLIKE (0.163), with Chronos-Bolt-Small close behind (0.174), while Moirai 2.0 underperforms substantially (QLIKE $= 0.329$, $R^2_{\text{OOS}} = -0.008$). At $h = 5$, Log-HAR retains the QLIKE lead (0.120 vs. 0.128 for Chronos-Bolt-Small), but Chronos-Bolt achieves the best MSE and R^2_{OOS} (0.353). At $h = 22$, Log-HAR again leads on QLIKE (0.147), while Chronos-Bolt achieves the best R^2_{OOS} (0.250). Moirai 2.0's poor FX performance ($R^2_{\text{OOS}} < 0$ at all horizons) stands in contrast to its equity dominance and suggests that Moirai's training data or architecture is less suited to the smoother, lower-amplitude dynamics of currency volatility.

Futures. The futures sample displays the widest cross-asset heterogeneity in volatility dynamics. The E-mini S&P 500 (ES) has first-order autocorrelation of 0.779, while Gold (GC) shows near-zero autocorrelation ($\rho_1 = -0.001$). At $h = 1$, Log-HAR achieves the lowest

QLIKE (0.193), followed by Chronos-Bolt-Small (0.213) and HAR (0.225). Moirai 2.0 ranks behind (0.289). At $h = 5$, Log-HAR retains the QLIKE lead (0.165), with Chronos-Bolt-Small at 0.185 and Chronos-Bolt-Base at 0.196. The TSFMs achieve the best MSE and R^2_{OOS}. At the monthly horizon, Moirai 2.0 recovers and achieves the lowest QLIKE (0.238), MSE, and MAE, while levels-based HAR variants produce inflated QLIKE (HAR: 1.08, HARQ: 17.88).

Cross-asset patterns. Three patterns emerge from the cross-asset comparison. First, Log-HAR is consistently the best or near-best econometric specification across all asset classes, confirming the practical value of the log transformation for ensuring forecast positivity and aligning with QLIKE's loss structure. Second, the relative ranking of TSFMs shifts across asset classes: Moirai 2.0 dominates equities (where persistence is high), while Chronos-Bolt performs better for FX and short-horizon futures (where dynamics are smoother or less persistent). Third, all levels-based HAR extensions produce inflated QLIKE for futures, where extreme volatility events in Natural Gas and Crude Oil expose the negative-forecast problem. These asset-class-dependent results highlight the importance of multi-market evaluations: a model that appears dominant on equities alone may not generalize to other asset classes.

5.3 Statistical Significance and Model Confidence Sets

Table 3 reports Model Confidence Set inclusion rates for 40 equities, and Table 6 summarizes pairwise Diebold–Mariano test results across all 50 assets.

MCS results for equities. At $h = 1$, Log-HAR enters the MCS for 90% of equities, confirming its status as the single most reliable econometric specification. Moirai 2.0 enters for 77.5%, HAR for 45%, and Chronos-Bolt-Small for 42.5%. Levels-based HAR variants are largely excluded: HAR-RS enters for only 5% and HARQ for 2.5%, reflecting their inflated QLIKE from negative forecasts.

At $h = 5$, the picture shifts. Moirai 2.0 enters the MCS for 100% of equities—every single stock—indicating that its forecasts are statistically indistinguishable from the best model for all assets at this horizon. Log-HAR enters for 72.5%, HAR for 50%, and HAR-J for 45%. Chronos-Bolt-Small enters for 30%, less than the baseline HAR. The econometric models' inclusion rates increase relative to $h = 1$, partly because forecast differentiation diminishes at longer horizons.

At $h = 22$, Moirai 2.0 enters for 95%, ARFIMA for 85%, and Log-HAR for 80%. The high ARFIMA inclusion rate at $h = 22$ is consistent with its long-memory parameterization: at the monthly horizon, fractional differencing captures slow mean reversion that the HAR's three fixed lags approximate less precisely. Chronos-Bolt-Small enters for 37.5%, and the levels-based HAR variants recover to 37.5–47.5%, as the MCS procedure becomes less punitive when overall forecast differentiation is low.

Diebold–Mariano pairwise tests. Table 6 reports the fraction of pairwise DM tests in which each model achieves significantly lower QLIKE at the 5% level, aggregated across 50 assets. At $h = 1$, ARFIMA wins the highest fraction of pairwise comparisons (80.8%), followed by Moirai 2.0 (66.2%) and HARQ (65.5%); Log-HAR ranks fourth at 63.7%. At $h = 5$ and $h = 22$, Moirai 2.0 achieves the highest win rate (56.5% and 29.5%, respectively), reflecting its broad dominance at longer horizons. The levels-based HAR variants (HAR-RS, HARQ) have the lowest DM win rates at all horizons, as they are significantly outperformed by most other models for a majority of assets.

Interpretation. The MCS and DM test results jointly indicate that zero-shot TSFMs have achieved statistical parity with the best econometric benchmarks for realized volatility forecasting. At $h = 1$, Log-HAR remains the marginal leader on QLIKE, but the TSFMs are statistically indistinguishable from it for most assets. At $h = 5$ and $h = 22$, Moirai 2.0 is the single model most consistently in the confidence set, and it outperforms econometric benchmarks in the majority of pairwise comparisons. The practical implication is that

TSFMs offer a viable alternative to HAR-family models, particularly for horizons beyond one day, and their zero-shot deployment requires no estimation, retraining, or feature engineering.

5.4 Portfolio Application: Realized Covariance Forecasting

The preceding analysis focused on univariate realized variance forecasting. We now extend the evaluation to a multivariate setting by examining whether covariance forecasts from TSFMs can improve portfolio allocation. We construct global minimum variance (GMV) portfolios using four approaches to covariance forecasting and compare the resulting realized portfolio variance.

Covariance forecasting methods. We consider four covariance forecasting approaches. (i) *Element-HAR*: a separate HAR model is fitted to each element of the realized covariance matrix, treating variances and covariances as independent univariate series. (ii) *HAR-DRD*: the realized covariance matrix is decomposed as $\Sigma_t = D_t R_t D_t$, where D_t is the diagonal matrix of realized volatilities and R_t is the realized correlation matrix; HAR models are fitted separately to each element of D_t and R_t, and the covariance forecast is reconstructed. This DRD decomposition ensures positive semi-definiteness by construction when the forecast correlation matrix is adjusted to be PSD. (iii–iv) *Chronos-Bolt-Small* and *Moirai 2.0*: zero-shot element-wise TSFM forecasts applied to each covariance pair, followed by PSD projection. We also include the $1/N$ equal-weight portfolio as a naive benchmark.

Forex GMV results. Table 7 reports the average realized variance of GMV portfolios constructed from each forecasting model, along with portfolio turnover and concentration metrics. For the 5-asset FX portfolio, all model-based approaches achieve substantial variance reduction relative to $1/N$. At $h = 1$, Chronos-Bolt-Small achieves the lowest average realized variance (3.01×10^{-6}), a 44.9% reduction relative to $1/N$ (5.46×10^{-6}). HAR-DRD is close behind (3.11×10^{-6}, 43.0% reduction), while Element-HAR (3.52×10^{-6}, 35.5%) and Moirai 2.0 (3.55×10^{-6}, 35.0%) are somewhat less effective. At $h = 5$, Chronos-Bolt

and HAR-DRD maintain their advantage (45.5% and 44.7% reduction, respectively), while Moirai 2.0 produces an unstable covariance forecast that results in higher realized variance than $1/N$. At $h = 22$, all methods except Moirai 2.0 produce comparable variance reductions (31–39%).

Futures GMV results. For the 5-contract futures portfolio (Table 7), Chronos-Bolt-Small achieves the lowest realized variance at $h = 1$ (3.36×10^{-5}, a 68.8% reduction vs. $1/N$) and $h = 5$ (3.55×10^{-5}, 67.2%). Moirai 2.0 performs worse, particularly at $h = 5$, where its realized variance exceeds $1/N$ by a factor of 3.5. This failure stems from Moirai's element-wise covariance forecasts being internally inconsistent: the assembled matrices are not positive semi-definite, and the eigenvalue floor used in PSD projection (10^{-10}) is too small relative to the matrix scale (typical eigenvalues $\sim 10^{-5}$). The resulting near-singular matrices produce extreme portfolio weights (max $|w| > 50$) on a handful of dates, which dominate the average realized variance. Chronos-Bolt does not suffer this problem because its forecasts produce matrices with minimum eigenvalues around 10^{-6}, keeping condition numbers below 10. Baseline covariance results for futures are pending and will be added when available.

Turnover. The $1/N$ portfolio has zero turnover by construction. Among model-based portfolios, HAR-DRD achieves the lowest turnover (0.080 for FX at $h = 1$), followed by Moirai 2.0 (0.060) and Chronos-Bolt (0.116). The DRD decomposition's smoothness advantage comes from its separate modeling of volatilities and correlations, which dampens spurious variation in off-diagonal elements. Chronos-Bolt's higher turnover reflects its more responsive covariance updates, which reduce portfolio variance at the cost of more frequent rebalancing.

Table 7: Global minimum variance portfolio performance. Avg RV is the average daily realized variance of the GMV portfolio. Turnover is the average daily absolute weight change. Max Wt is the average maximum portfolio weight.

Model	h	Avg RV ($\times 10^{-6}$)	Std RV ($\times 10^{-6}$)	Turnover	Max Wt
Forex (5 assets)					
$1/N$	1	5.46	10.30	—	—
Element-HAR		3.52	6.50	0.096	0.383
HAR-DRD		3.11	4.50	0.080	0.370
Chronos-Bolt-S		**3.01**	4.78	0.116	0.369
Moirai-2.0-S		3.55	9.83	0.060	0.384
$1/N$	5	5.73	6.48	—	—
Element-HAR		3.58	5.45	0.096	0.383
HAR-DRD		3.17	2.99	0.080	0.370
Chronos-Bolt-S		**3.12**	3.69	0.090	0.369
Moirai-2.0-S		21.72	1083.82	0.104	0.397
$1/N$	22	5.43	4.32	—	—
Element-HAR		3.71	5.89	0.096	0.383
HAR-DRD		3.30	2.45	0.080	0.370
Chronos-Bolt-S		**3.29**	3.00	0.074	0.370
Moirai-2.0-S		3.52	6.06	0.046	0.382
Futures (5 contracts)					
$1/N$	1	107.77	316.84	—	—
Chronos-Bolt-S		**33.63**	33.30	0.114	0.520
Moirai-2.0-S		45.08	156.03	0.264	0.552
$1/N$	5	107.88	207.81	—	—
Chronos-Bolt-S		**35.46**	32.45	0.088	0.514
Moirai-2.0-S		376.26	19571.31	0.321	0.560
$1/N$	22	108.35	151.33	—	—
Chronos-Bolt-S		**39.88**	44.59	0.065	0.503
Moirai-2.0-S		59.63	670.30	0.154	0.515

Table 8: Covariance forecast accuracy for 5 FX pairs. Avg Frobenius is the average Frobenius norm of the forecast error matrix. Diag QLIKE is the average QLIKE computed on diagonal (variance) elements.

Model	h	Avg Frobenius ($\times 10^{-5}$)	Diag QLIKE
Element-HAR	1	1.057	0.149
HAR-DRD		0.960	0.133
Element-HAR	5	0.891	0.105
HAR-DRD		0.789	0.092
Element-HAR	22	0.878	0.116
HAR-DRD		0.789	0.112

6 Robustness Checks

We assess the stability of our main findings along two dimensions: cross-dataset validation and the behavior of model rankings across the full set of results.

Cross-dataset validation. We replicate the univariate forecasting exercise using the CAPIRe dataset, which provides realized volatility for 29 DJIA stocks over 2003–2024 in annualized percentage units—a different scale, sample period, and estimation methodology than VOLARE. Tables 9 and 10 (Appendix) report the results.

The CAPIRe results confirm the main patterns observed with VOLARE. TSFMs achieve the lowest MSE and highest R^2_{OOS} at all horizons: Chronos-Bolt-Small achieves $R^2_{\text{OOS}} = 0.465$ at $h = 1$ and 0.495 at $h = 5$, compared to 0.382 and 0.369 for Log-HAR. On QLIKE, Log-HAR is best at $h = 1$ (0.139 vs. 0.149 for Chronos-Bolt-Small), but at $h = 5$ and $h = 22$ Chronos-Bolt-Small achieves the lowest QLIKE (0.155 and 0.249, respectively). Model rankings between VOLARE and CAPIRe are broadly consistent, with Log-HAR and the TSFMs consistently at the top and levels-based HAR variants penalized by inflated QLIKE. The stability of these findings across two independent datasets with different time periods, asset universes, and RV estimation methodologies strengthens the conclusion that

TSFMs provide competitive zero-shot performance.

Sub-sample stability: pre- and post-COVID. Table **??** splits the equity sample at March 1, 2020, and reports forecast accuracy separately for the pre-COVID (2015–2020) and post-COVID (2020–2026) periods. The sub-sample analysis reveals that TSFM performance is regime-robust, while levels-based econometric models are not.

In the pre-COVID period, Log-HAR achieves the lowest QLIKE at all horizons (0.206 at $h = 1$, 0.182 at $h = 5$, 0.321 at $h = 22$), though Chronos-Bolt-Base matches it on MSE and R^2_{OOS} at $h = 1$. The pre-COVID period features lower average volatility and fewer extreme events, conditions that favor the parsimonious HAR specification.

In the post-COVID period, TSFMs dominate across all metrics and horizons. Moirai 2.0 achieves the lowest QLIKE at $h = 1$ (0.213 vs. 0.225 for Log-HAR), $h = 5$ (0.174 vs. 0.252), and $h = 22$ (0.168 vs. 0.335), while also achieving the highest R^2_{OOS} (0.430, 0.378, 0.148). The levels-based HAR variants suffer extreme QLIKE inflation in the high-volatility regime: HARQ's QLIKE rises from 2.35 pre-COVID to 23.63 post-COVID at $h = 1$, because the near-zero forecast problem is exacerbated when realized values are large. Log-HAR is more stable across regimes than its levels-based counterparts but still falls behind the TSFMs on all metrics in the post-COVID sample. These results suggest that TSFMs' advantage is concentrated in volatile market conditions, where their flexible nonparametric forecasts adapt more effectively than the HAR's linear structure.

Remaining directions. VOLARE provides alternative realized measures including sub-sampled realized variance ($RV^{(5)}_{\text{SS}}$), realized kernels, and range-based estimators. Evaluating model rankings under alternative volatility proxies and varying TSFM context lengths (128 to 512 days) are directions for future work that would further establish the generality of our findings.

Table 9: Sub-sample forecast accuracy: pre-COVID (2015–2020) and post-COVID (2020–2026) periods for 40 U.S. equities (VOLARE).

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$, Pre-COVID				
HAR	0.089	0.984	0.494	0.223
HAR-J	0.090	0.976	0.395	0.204
HAR-RS	0.102	1.037	1.312	0.158
HARQ	0.091	0.959	2.346	0.175
Log-HAR	0.087	0.862	**0.206**	0.242
ARFIMA	0.123	1.148	0.328	-0.008
Chronos-Bolt-S	0.086	0.834	0.235	0.247
Chronos-Bolt-B	**0.084**	**0.823**	0.234	**0.275**
Moirai-2.0-S	0.084	0.857	0.251	0.235
Panel: $h = 1$, Post-COVID				
HAR	0.328	1.684	1.809	0.330
HAR-J	0.324	1.667	1.539	0.345
HAR-RS	0.356	1.797	5.151	0.259
HARQ	0.406	1.780	23.625	0.091
Log-HAR	0.348	1.545	0.225	0.225
ARFIMA	0.392	1.896	0.312	0.147
Chronos-Bolt-S	0.285	1.363	0.226	0.410
Chronos-Bolt-B	0.281	**1.353**	0.228	0.422
Moirai-2.0-S	**0.277**	1.360	**0.213**	**0.430**
Panel: $h = 5$, Pre-COVID				
HAR	0.050	0.972	0.421	0.029
HAR-J	0.052	0.966	0.427	0.018
HAR-RS	0.061	1.025	2.051	-0.098
HARQ	0.050	0.954	1.814	0.036
Log-HAR	**0.044**	0.839	**0.182**	0.190
ARFIMA	0.063	1.014	0.263	-0.063
Chronos-Bolt-S	0.047	0.791	0.241	0.168
Chronos-Bolt-B	0.046	**0.782**	0.240	0.187
Moirai-2.0-S	0.045	0.789	0.217	**0.202**
Panel: $h = 5$, Post-COVID				
HAR	0.221	1.643	3.877	0.193
HAR-J	0.214	1.621	3.529	0.219
HAR-RS	0.249	1.777	9.622	0.077
HARQ	0.265	1.694	23.517	-0.008
Log-HAR	0.226	1.500	0.252	0.108
ARFIMA	0.227	1.672	0.237	0.167
Chronos-Bolt-S	0.170	1.256	0.200	0.348
Chronos-Bolt-B	0.172	1.261	0.204	0.337
Moirai-2.0-S	**0.162**	**1.216**	**0.174**	**0.378**
Panel: $h = 22$, Pre-COVID				
HAR	0.095	1.145	1.587	-0.032
HAR-J	0.095	1.140	1.510	-0.028
HAR-RS	0.101	1.197	3.796	-0.084

7 Conclusion

This paper evaluates zero-shot time series foundation models for realized volatility forecasting across 50 assets spanning U.S. equities, foreign exchange, and futures. We compare two TSFMs—Chronos-Bolt and Moirai 2.0—against six econometric benchmarks at three forecast horizons using the VOLARE dataset, with formal statistical testing via Diebold–Mariano tests and Model Confidence Sets.

Three main findings emerge. First, zero-shot TSFMs are competitive with the best econometric benchmarks. This result differs from prior work on daily returns (Rahimikia et al., 2025) and on TimesFM 1.0 (Goel et al., 2025), and is consistent with the hypothesis that realized volatility's positive, persistent, mean-reverting dynamics align more closely with the macroeconomic and physical time series on which TSFMs were pretrained. Second, the ranking of models depends on the loss function and forecast horizon. Log-HAR achieves the lowest QLIKE at $h = 1$ for equities, but TSFMs dominate on MSE, MAE, and R^2_{OOS} at all horizons. At $h = 5$ and $h = 22$, Moirai 2.0 achieves the lowest QLIKE as well. Model Confidence Set analysis shows that Moirai 2.0 enters the MCS for 100% of equities at $h = 5$ and 95% at $h = 22$, while Log-HAR enters for 90% at $h = 1$. Third, model rankings are not stable across asset classes: Chronos-Bolt outperforms on FX and futures, while Moirai 2.0 dominates equities.

A practical finding is that levels-based HAR extensions (HAR-RS, HARQ) produce near-zero forecasts that inflate QLIKE by orders of magnitude, a problem that log-space models and TSFMs avoid. Sub-sample analysis reveals that TSFM advantages are concentrated in the volatile post-COVID period (2020–2026), where Moirai 2.0 dominates all metrics and horizons, while Log-HAR retains a QLIKE edge in the calmer pre-COVID period. For portfolio construction, element-wise zero-shot TSFM forecasts of realized covariance matrices achieve substantial variance reductions (35–69%) relative to $1/N$ in GMV portfolios, with Chronos-Bolt performing comparably to HAR-DRD baselines.

Several limitations should be noted. We evaluate only zero-shot TSFM performance; fine-

tuning on realized volatility data may yield further gains. Our evaluation is restricted to point forecasts; density forecast evaluation and interval calibration (Adler et al., 2025) remain open questions. We use a single volatility estimator (5-minute realized variance); sensitivity to the choice of estimator is left for future work. The covariance forecasting exercise uses a simple element-wise approach; natively multivariate TSFM architectures (Ansari et al., 2025) may perform better.

Future work should pursue four directions. First, parameter-efficient fine-tuning (e.g., LoRA) on realized volatility data may close the remaining gap with Log-HAR on QLIKE at short horizons. Second, density forecast evaluation would test whether TSFMs provide well-calibrated prediction intervals, a property required for Value-at-Risk and Expected Shortfall applications. Third, natively multivariate TSFMs could be applied directly to realized covariance matrices, potentially outperforming the element-wise approach used here. Fourth, extending the evaluation to emerging markets and alternative asset classes would test the generality of our findings.

The broader implication of our results is that zero-shot time series foundation models have reached a level of accuracy that makes them a practical tool for volatility forecasting. For practitioners, TSFMs combine competitive accuracy with zero estimation overhead and immediate deployment to new assets—advantages that become increasingly valuable as the number of assets to be forecast grows.

A CAPIRe Dataset Results

Table 10: Forecast accuracy for 29 DJIA stocks (CAPIRe dataset). Values are cross-sectional averages. RV is in annualized % units. Bold indicates best value per column per panel. † denotes inflated QLIKE.

Model	MSE	MAE	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	24.059	1.08	0.474	0.383
HAR-J	24.213	1.08	0.512	0.373
HAR-RS	25.077	1.14	1.27†	0.353
HARQ	29.702	1.11	12.57†	0.247
Log-HAR	23.642	0.98	**0.139**	0.382
ARFIMA	27.555	1.29	0.276	0.281
Chronos-Bolt-S	21.150	0.91	0.149	0.465
Chronos-Bolt-B	**20.940**	0.90	0.148	**0.466**
Moirai-2.0-S	21.537	**0.89**	0.233	0.452
Panel: $h = 5$ (5 days)				
HAR	14.718	1.14	1.28†	0.371
HAR-J	14.883	1.13	1.28†	0.364
HAR-RS	15.454	1.21	3.25†	0.323
HARQ	18.231	1.16	13.26†	0.230
Log-HAR	14.416	1.00	0.156	0.369
ARFIMA	15.626	1.19	0.252	0.306
Chronos-Bolt-S	**11.932**	0.89	**0.155**	**0.495**
Chronos-Bolt-B	12.142	**0.88**	0.156	0.483
Moirai-2.0-S	12.868	0.88	0.214	0.451
Panel: $h = 22$ (22 days)				
HAR	13.853	1.40	5.29†	0.085
HAR-J	14.051	1.40	5.25†	0.070
HAR-RS	15.030	1.48	7.82†	-0.025
HARQ	318.153	1.49	12.88†	-12.386
Log-HAR	12.829	1.23	0.260	0.164
ARFIMA	12.789	1.20	0.286	0.090
Chronos-Bolt-S	**10.846**	1.01	**0.249**	**0.308**
Chronos-Bolt-B	11.069	1.00	0.255	0.289
Moirai-2.0-S	11.327	**0.99**	0.292	0.270

Table 11: Model Confidence Set inclusion rates for 29 DJIA stocks (CAPIRe dataset). Format as Table 3.

Model	$h = 1$ (%)	$h = 5$ (%)	$h = 22$ (%)
HAR	3.4	44.8	86.2
HAR-J	6.9	48.3	86.2
HAR-RS	3.4	27.6	72.4
HARQ	0.0	27.6	75.9
Log-HAR	96.6	96.6	100.0
ARFIMA	0.0	27.6	93.1
Chronos-Bolt-S	44.8	86.2	93.1
Chronos-Bolt-B	58.6	86.2	93.1
Moirai-2.0-S	51.7	75.9	93.1

B Median Aggregation

Table **??** reports cross-sectional medians rather than means for the 40-equity sample, providing robustness to outlier assets with degenerate forecasts (e.g., HARQ at $h = 22$).

Table 12: Forecast accuracy for 40 U.S. equities (VOLARE), cross-sectional medians. Median aggregation is robust to outlier assets with degenerate forecasts. Bold indicates the best value in each column within each panel.

Model	MSE ($\times 10^{-6}$)	MAE ($\times 10^{-4}$)	QLIKE	R^2_{OOS}
Panel: $h = 1$ (1 day)				
HAR	0.116	1.14	0.221	0.352
HAR-J	0.113	1.14	0.567	0.365
HAR-RS	0.124	1.20	$1.37^{\dagger}$	0.318
HARQ	0.173	1.24	$11.12^{\dagger}$	0.066
Log-HAR	0.139	1.12	**0.208**	0.237
ARFIMA	0.147	1.37	0.297	0.163
Chronos-Bolt-S	0.096	0.97	0.212	0.436
Chronos-Bolt-B	0.094	**0.95**	0.213	**0.459**
Moirai-2.0-S	**0.091**	0.98	0.208	0.451
Panel: $h = 5$ (5 days)				
HAR	0.096	1.13	0.269	0.253
HAR-J	0.094	1.12	0.384	0.271
HAR-RS	0.109	1.19	$3.59^{\dagger}$	0.179
HARQ	0.126	1.17	$13.39^{\dagger}$	0.017
Log-HAR	0.111	1.08	0.205	0.133
ARFIMA	0.098	1.19	0.232	0.192
Chronos-Bolt-S	0.077	0.90	0.196	0.366
Chronos-Bolt-B	0.080	0.91	0.196	0.359
Moirai-2.0-S	**0.073**	**0.86**	**0.172**	**0.383**
Panel: $h = 22$ (22 days)				
HAR	0.078	1.23	$3.40^{\dagger}$	0.007
HAR-J	0.076	1.22	$3.39^{\dagger}$	0.016
HAR-RS	0.082	1.30	$5.23^{\dagger}$	-0.090
HARQ	0.089	1.25	$9.29^{\dagger}$	-0.121
Log-HAR	0.081	1.13	0.296	-0.043
ARFIMA	0.075	1.10	0.269	0.073
Chronos-Bolt-S	0.071	0.94	0.319	0.085
Chronos-Bolt-B	0.072	0.97	0.326	0.082
Moirai-2.0-S	**0.069**	**0.89**	**0.248**	**0.112**

References

Coen Adler, Yuxin Chang, Felix Draxler, Samar Abdi, and Padhraic Smyth. Beyond accuracy: Are time series foundation models well-calibrated? *arXiv preprint arXiv:2510.16060*, 2025.

Taha Aksu, Gerald Woo, Juncheng Liu, Xu Liu, Chenghao Liu, Silvio Savarese, Caiming Xiong, and Doyen Sahoo. GIFT-Eval: A benchmark for general time series forecasting model evaluation. *arXiv preprint arXiv:2410.10393*, 2024.

Torben G. Andersen and Tim Bollerslev. Answering the skeptics: Yes, standard volatility models do provide accurate forecasts. *International Economic Review*, 39(4):885–905, 1998. doi: 10.2307/2527343.

Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. The distribution of realized exchange rate volatility. *Journal of the American Statistical Association*, 96 (453):42–55, 2001. doi: 10.1198/016214501750332965.

Torben G. Andersen, Tim Bollerslev, Francis X. Diebold, and Paul Labys. Modeling and forecasting realized volatility. *Econometrica*, 71(2):579–625, 2003. doi: 10.1111/1468-0262.00418.

Torben G. Andersen, Tim Bollerslev, and Francis X. Diebold. Roughing it up: Including jump components in the measurement, modeling, and forecasting of return volatility. *Review of Economics and Statistics*, 89(4):701–720, 2007. doi: 10.1162/rest.89.4.701.

Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024. arXiv:2403.07815.

Abdul Fatir Ansari, Oleksandr Shchur, Jaris Küken, Andreas Auer, Boran Han, Pedro Mercado, Syama Sundar Rangapuram, Huibin Shen, Lorenzo Stella, Xiyuan Zhang, Mononito Goswami, Shubham Kapoor, Danielle C. Maddix, Pablo Guerron, Tony Hu, Junming Yin, Nick Erickson, Prateek Mutalik Desai, Hao Wang, Huzefa Rangwala, George Karypis, Yuyang Wang, and Michael Bohlke-Schneider. Chronos-2: From univariate to universal forecasting. *arXiv preprint arXiv:2510.15821*, 2025.

Richard T. Baillie, Tim Bollerslev, and Hans Ole Mikkelsen. Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74(1):3–30, 1996. doi: 10.1016/S0304-4076(95)01749-6.

Ole E. Barndorff-Nielsen and Neil Shephard. Econometric analysis of realized volatility and its use in estimating stochastic volatility models. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 64(2):253–280, 2002. doi: 10.1111/1467-9868.00336.

Ole E. Barndorff-Nielsen and Neil Shephard. Power and bipower variation with stochastic volatility and jumps. *Journal of Financial Econometrics*, 2(1):1–37, 2004. doi: 10.1093/jjfinec/nbh001.

Ole E. Barndorff-Nielsen, Peter Reinhard Hansen, Asger Lunde, and Neil Shephard. Designing realized kernels to measure the ex post variation of equity prices in the presence of noise. *Econometrica*, 76(6):1481–1536, 2008. doi: 10.3982/ECTA6495.

Ole E. Barndorff-Nielsen, Silja Kinnebrock, and Neil Shephard. Measuring downside risk: Realised semivariance. In Tim Bollerslev, Jeffrey Russell, and Mark Watson, editors, *Volatility and Time Series Econometrics: Essays in Honor of Robert F. Engle*, pages 117–136. Oxford University Press, 2010.

Mikkel Bennedsen, Asger Lunde, and Mikko S. Pakkanen. Decoupling the short- and long-term behavior of stochastic volatility. *Journal of Financial Econometrics*, 20(5):961–1006, 2022. doi: 10.1093/jjfinec/nbaa049.

Tim Bollerslev, Andrew J. Patton, and Rogier Quaedvlieg. Exploiting the errors: A simple approach for improved volatility forecasting. *Journal of Econometrics*, 192(1):1–18, 2016. doi: 10.1016/j.jeconom.2015.10.007.

Rafael Branco, Alexandre Rubesam, and Mauricio Zevallos. Forecasting realized volatility: Does anything beat linear models? *Journal of Empirical Finance*, 78:101524, 2024. doi: 10.1016/j.jempfin.2024.101524.

Andrea Bucci. Realized volatility forecasting with neural networks. *Journal of Financial Econometrics*, 18(3):502–531, 2020. doi: 10.1093/jjfinec/nbaa008.

Kim Christensen, Mathias Siggaard, and Bezirgen Veliyev. A machine learning approach to volatility forecasting. *Journal of Financial Econometrics*, 21(5):1680–1727, 2023. doi: 10.1093/jjfinec/nbac020.

Fabrizio Cipollini, Giulia Cruciani, Giampiero M. Gallo, Alessandra Insana, Edoardo Otranto, and Fabio Spagnolo. VOLARE: VOLatility Archive for Realized Estimates. *arXiv preprint arXiv:2602.19732*, 2026.

Adam Clements and Daniel P. A. Preve. A practical guide to harnessing the HAR volatility model. *Journal of Banking & Finance*, 133:106285, 2021. doi: 10.1016/j.jbankfin.2021.106285.

Fulvio Corsi. A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7(2):174–196, 2009. doi: 10.1093/jjfinec/nbp001.

Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 10148–10167. PMLR, 2024. arXiv:2310.10688.

Abhimanyu Das, Matthew Faw, Rajat Sen, and Yichen Zhou. In-context fine-tuning for time-series foundation models. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2025. arXiv:2410.24087.

Francis X. Diebold and Roberto S. Mariano. Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3):253–263, 1995. doi: 10.1080/07350015.1995.10524599.

Jim Gatheral, Thibault Jaisson, and Mathieu Rosenbaum. Volatility is rough. *Quantitative Finance*, 18(6):933–949, 2018. doi: 10.1080/14697688.2017.1393551.

M. Germán-Morales, A. J. Rivera-Rivas, M. J. del Jesus Díaz, and C. J. Carmona. Transfer learning with foundational models for time series forecasting using low-rank adaptations. *arXiv preprint arXiv:2410.11539*, 2024.

Anubha Goel, Puneet Pasricha, and Juho Kanniainen. Time-series foundation AI model for value-at-risk forecasting. *arXiv preprint arXiv:2410.11773*, 2024.

Anubha Goel, Puneet Pasricha, Martin Magris, and Juho Kanniainen. Foundation time-series AI model for realized volatility forecasting. *arXiv preprint arXiv:2505.11163*, 2025.

Mononito Goswami, Konrad Szafer, Arjun Choudhry, Yifu Cai, Shuo Li, and Artur Dubrawski. MOMENT: A family of open time-series foundation models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*. PMLR, 2024. arXiv:2402.03885.

Elias Søvik Gunnarsson, Håkon Ramon Isern, Aristidis Kaloudis, Morten Risstad, Benjamin Vigdel, and Sjur Westgaard. Prediction of realized volatility and implied volatility indices using AI and machine learning: A review. *International Review of Financial Analysis*, 93: 103221, 2024. doi: 10.1016/j.irfa.2024.103221.

Divij Gupta, Anubhav Bhatti, and Surajsinh Parmar. Beyond LoRA: Exploring efficient fine-

tuning techniques for time series foundational models. *arXiv preprint arXiv:2409.11302*, 2024.

Peter Reinhard Hansen, Asger Lunde, and James M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011. doi: 10.3982/ECTA5771.

Marwan Izzeldin, M. Kabir Hassan, Vasileios Pappas, and Mike Tsionas. Forecasting realised volatility using ARFIMA and HAR models. *Quantitative Finance*, 19(10):1627–1638, 2019. doi: 10.1080/14697688.2019.1600713.

Stanisław Łaniewski and Robert Ślepaczuk. Evaluating the Chronos foundation model for daily stock index forecasting. In *Proceedings of the International Conference on Information Systems Development (ISD 2025)*, 2025. doi: 10.62036/ISD.2025.48.

Radmir Mishelevich Leushuis and Nicolai Petkov. Advances in forecasting realized volatility: A review of methodologies. *Financial Innovation*, 12:5, 2026. doi: 10.1186/s40854-025-00809-5.

Zhe Li, Xiangfei Qiu, Peng Chen, Yihang Wang, Hanyin Cheng, Yang Shu, Jilin Hu, Chenjuan Guo, Aoying Zhou, Christian S. Jensen, and Bin Yang. TSFM-Bench: A comprehensive and unified benchmark of foundation models for time series forecasting. *arXiv preprint arXiv:2410.11802*, 2024. KDD 2025.

Yuxuan Liang, Haomin Wen, Yuqi Nie, Yushan Jiang, Ming Jin, Dongjin Song, Shirui Pan, and Qingsong Wen. Foundation models for time series analysis: A tutorial and survey. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2024. arXiv:2403.14735.

Chenghao Liu, Taha Aksu, Juncheng Liu, Xu Liu, Hanshu Yan, Quang Pham, Silvio Savarese, Doyen Sahoo, Caiming Xiong, and Junnan Li. Moirai 2.0: When less is more for time series forecasting. *arXiv preprint arXiv:2511.11698*, 2025a.

Yong Liu, Haoran Zhang, Chenyu Li, Xiangdong Huang, Jianmin Wang, and Mingsheng Long. Timer: Generative pre-trained transformers are large time series models. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*. PMLR, 2024. arXiv:2402.02368.

Yong Liu, Guo Qin, Zhiyuan Shi, Zhi Chen, Caiyin Yang, Xiangdong Huang, Jianmin Wang, and Mingsheng Long. Sundial: A family of highly capable time series foundation models. *arXiv preprint arXiv:2502.00816*, 2025b. ICML 2025 Oral (Top 1%).

Chuong Luong and Nikolai Dokuchaev. Forecasting of realised volatility with the random forests algorithm. *Journal of Risk and Financial Management*, 11(4):61, 2018. doi: 10.3390/jrfm11040061.

Ben A. Marconi. Time series foundation models for multivariate financial time series forecasting. *arXiv preprint arXiv:2507.07296*, 2025.

John A. Miller, Mohammed Aldosari, Farah Saeed, Nasid Habib Barna, Subas Rana, I. Budak Arpinar, and Ninghao Liu. A survey of deep learning and foundation models for time series forecasting. *arXiv preprint arXiv:2401.13912*, 2024.

Fernando Moreno-Pino and Stefan Zohren. DeepVol: Volatility forecasting from high-frequency data with dilated causal convolutions. *Quantitative Finance*, 24(8):1105–1127, 2024. doi: 10.1080/14697688.2024.2387222.

Quoc Anh Nguyen, Ce Guo, and Wayne Luk. Repurposing language models for FX volatility forecasting: A data-efficient and context-aware approach. In *Proceedings of the 6th ACM International Conference on AI in Finance (ICAIF '25)*. ACM, 2025. doi: 10.1145/3768292.3770386.

Andrew J. Patton. Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1):246–256, 2011. doi: 10.1016/j.jeconom.2010.03.034.

Andrew J. Patton and Kevin Sheppard. Good volatility, bad volatility: Signed jumps and the persistence of volatility. *Review of Economics and Statistics*, 97(3):683–697, 2015. doi: 10.1162/REST_a_00503.

Eghbal Rahimikia, Hao Ni, and Weiguan Wang. Re(Visiting) time series foundation models in finance. *arXiv preprint arXiv:2511.18578*, 2025.

Kashif Rasul, Arjun Ashok, Andrew Robert Williams, Hena Ghonia, Rishika Bhagwatkar, Arian Khorasani, Mohammad Javad Darvishi Bayazi, George Adamopoulos, Roland Riachi, Nadhir Hassen, Marin Biloš, Sahil Garg, Anderson Schneider, Nicolas Chapados, Alexandre Drouin, Valentina Zantedeschi, Yuriy Nevmyvaka, and Irina Rish. Lag-llama: Towards foundation models for probabilistic time series forecasting. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2024. arXiv:2310.08278.

Oleksandr Shchur, Abdul Fatir Ansari, Caner Turkmen, Lorenzo Stella, Nick Erickson, Pablo Guerron, Michael Bohlke-Schneider, and Yuyang Wang. fev-bench: A realistic benchmark for time series forecasting. *arXiv preprint arXiv:2509.26468*, 2025.

Yu Shi, Zongliang Fu, Shuo Chen, Bohan Zhao, Wei Xu, Changshui Zhang, and Jian Li. Kronos: A foundation model for the language of financial markets. *arXiv preprint arXiv:2508.02739*, 2025.

Sébastien Valeyre and Sofiane Aboura. LLMs for time series: An application for single stocks and statistical arbitrage. *arXiv preprint arXiv:2412.09394*, 2024.

Gerald Woo, Chenghao Liu, Akshat Kumar, Caiming Xiong, Silvio Savarese, and Doyen Sahoo. Unified training of universal time series forecasting transformers. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pages 53140–53164. PMLR, 2024. arXiv:2402.02592. Oral presentation.

Jiexia Ye, Yongzi Yu, Weiqi Zhang, Le Wang, Jia Li, and Fugee Tsung. Empowering time series analysis with foundation models: A comprehensive survey. *arXiv preprint arXiv:2405.02358*, 2024.

Zhuohang Zhu, Haodong Chen, Qiang Qu, and Vera Chung. FinCast: A foundation model for financial time-series forecasting. *arXiv preprint arXiv:2508.19609*, 2025. CIKM 2025.

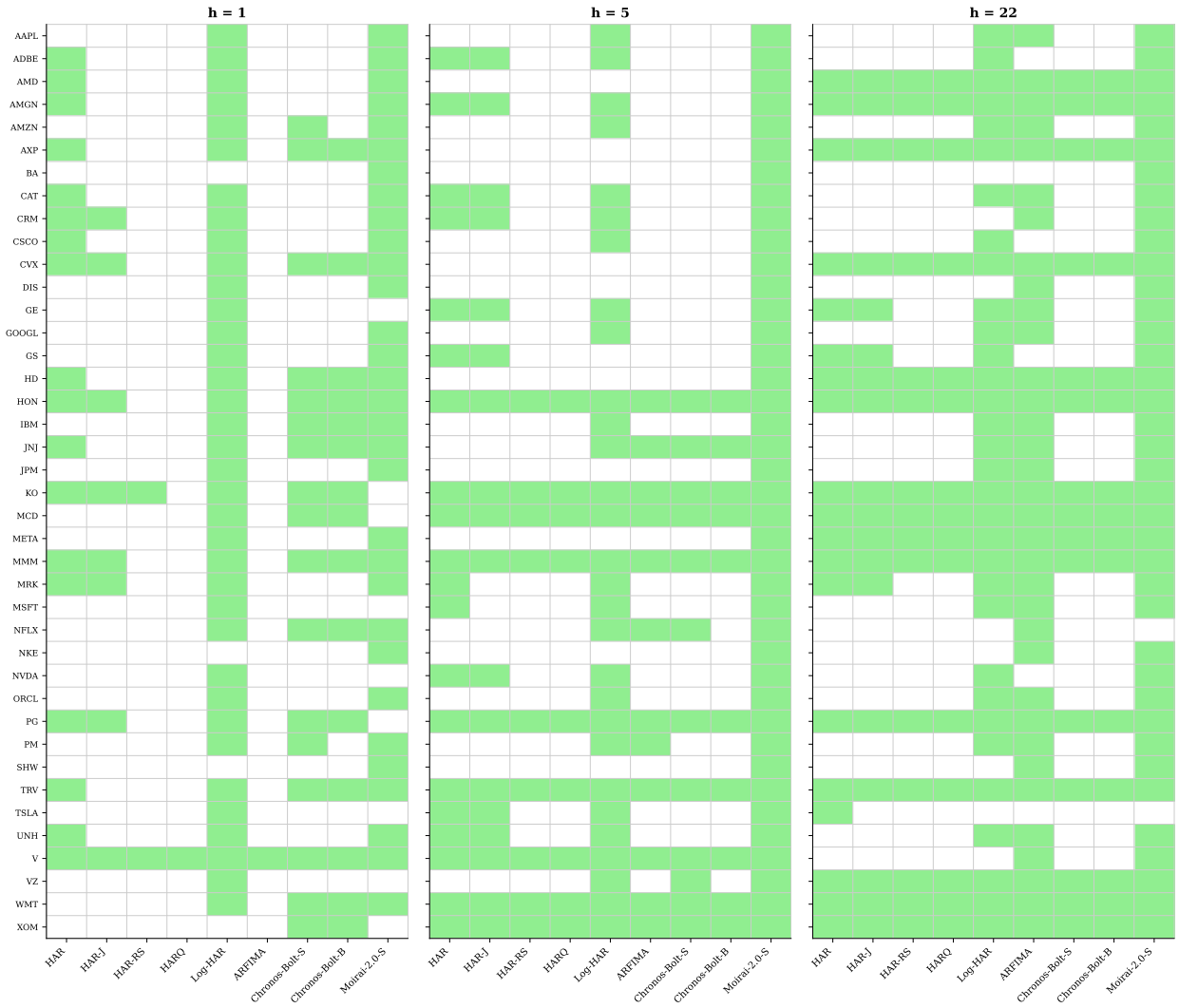

Figure 2: Model Confidence Set inclusion for 40 U.S. equities. Green cells indicate that the model is included in the MCS at the 10% significance level (QLIKE loss). Left: $h = 1$; center: $h = 5$; right: $h = 22$.

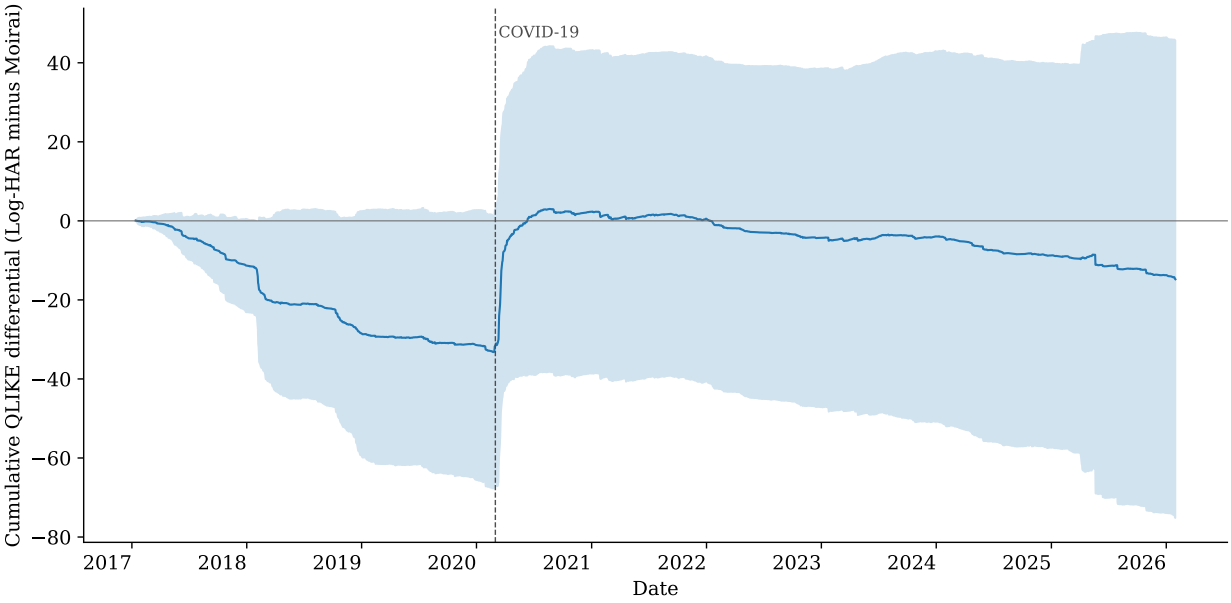

Figure 3: Cumulative QLIKE loss differential (Log-HAR minus Moirai 2.0) at $h = 1$, averaged across 40 equities. Positive values indicate that Moirai 2.0 outperforms Log-HAR. The shaded area is ± 1 cross-sectional standard deviation. The vertical dashed line marks March 2020 (COVID-19 onset).